

VETO



The Commit Gate for Autonomous Analytics

The first autonomous data analyst you can let run **unsupervised** — because an adversarial agent with veto power stands between every conclusion it reaches and every action it takes.

THE PROBLEM IT KILLS

Enterprises don't fear *dumb* agents.
They fear **confident, wrong** agents with
execution rights.

1

An agent acts

It pushes a flawed conclusion to a dashboard or a Teams channel — autonomously, with no human in the loop.

2

Someone ships it

A wrong automated insight becomes a real decision: a budget cut, a defunded team, a treatment choice.

3

Trust collapses

That single fear is the #1 blocker to deploying autonomous analytics at all.

LLMs are structurally unreliable at statistical inference — they hallucinate rigor. So we never ask them to do inference.

THE CORE IDEA

Separate generation from verification. Put a commit gate between them.



PROPOSE

An LLM Analyst reads the data and commits to an action with a victim — "cut Region B." Fast, decisive, naive.



VERIFY

An Adversary runs a deterministic statistical battery — scipy / statsmodels. Verdicts are computed, not claimed.



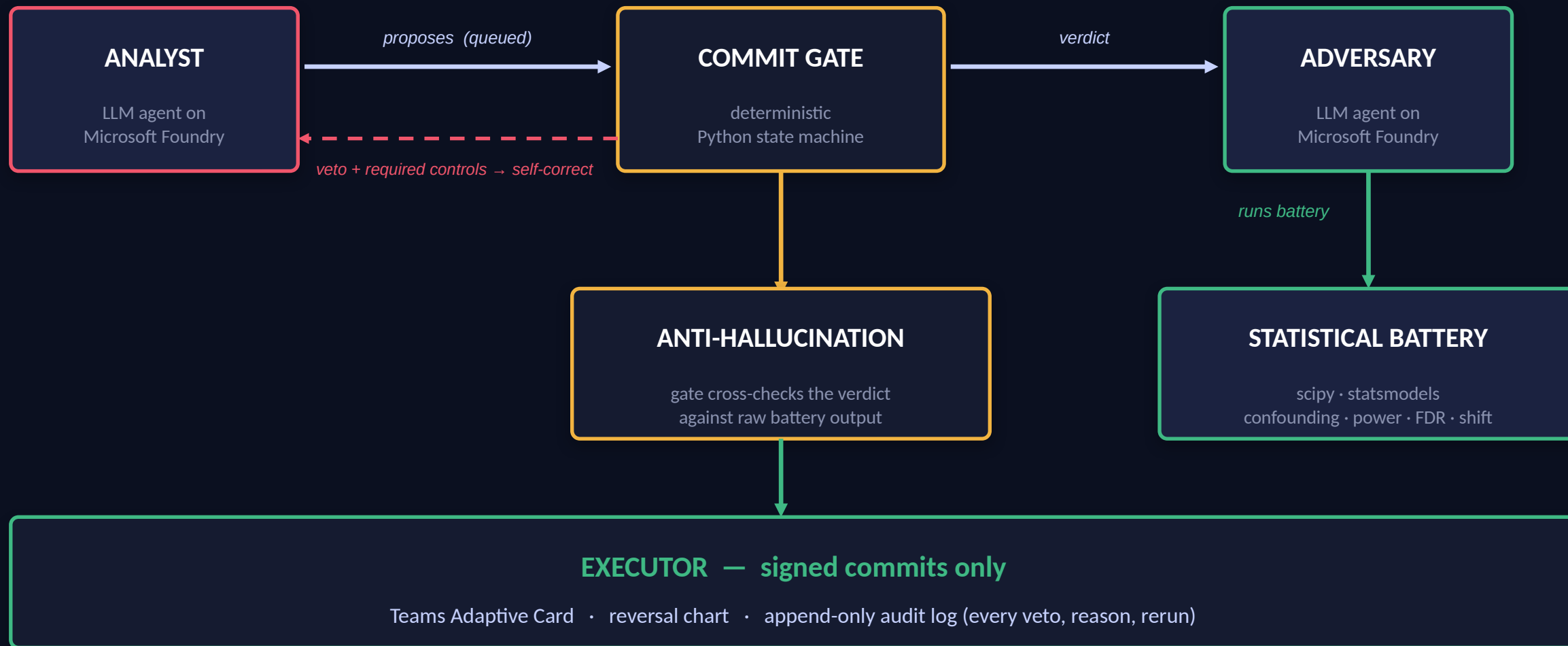
GATE

Pure-Python code holds the only execution keys. Nothing runs without a signed commit it can independently verify.

The authority comes from scipy — not from a second model agreeing with the first.

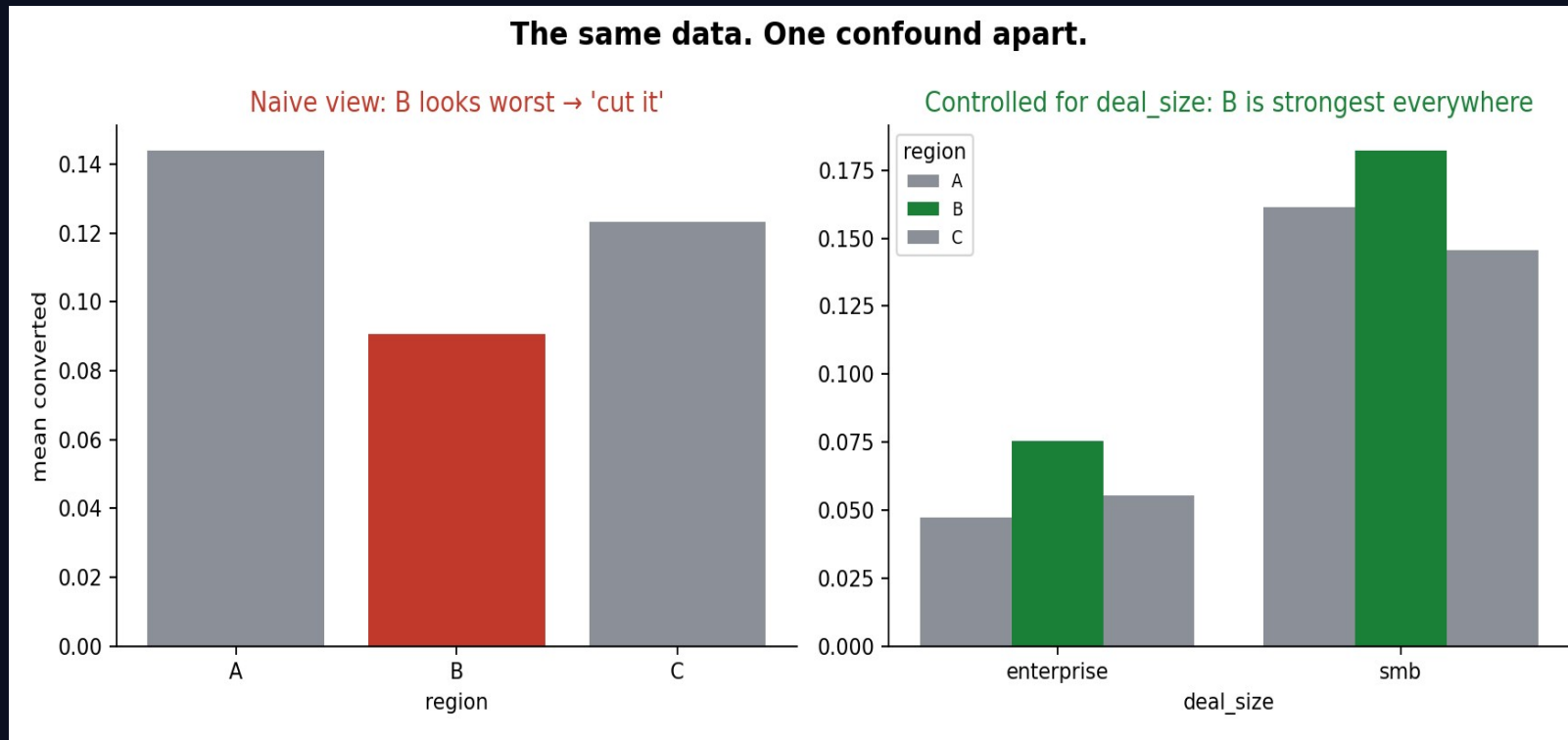
ARCHITECTURE

Agents propose and critique. Only deterministic code holds the keys.



THE GUT-PUNCH

The same data. One confound apart.



NAIVE

Region B looks worst.
"Cut it."

CONTROLLED

Once you account for deal size, B is the **strongest** region in every segment.

Approved as-is, this would have defunded the best team in the company.

The battery fires the confounding check — sign reversal, naive $-0.043 \rightarrow$ controlled $+0.025$ ($p = 0.002$) — and vetoes the commit.

WHY IT'S NOT A ONE-TRICK DEMO

Different data. Different traps. Same gate.



Confounding

Simpson's reversal

Stratified OLS: the effect flips sign once you control for the covariate.

headline_sales.csv



Statistical power

Underpowered "trend"

Achieved power 0.27, bootstrap CI crosses zero — the gap isn't real.

trap_power.csv



Multiplicity

p-hacking bait

20-metric scan; the lone nominal hit dies under Benjamini-Hochberg FDR.

trap_multiplicity.csv

The seeded dataset generators ship in the repo — generate your own and try to sneak one past the gate.

WHY THIS IS HARD — AND MICROSOFT-NATIVE

Built so the model can fail and it doesn't matter.



The gate is code, not an agent

A deterministic state machine holds the only execution keys. The Adversary reports; the gate decides.



Anti-hallucination cross-check

The verdict must cite a battery run id; the gate independently holds the raw output and overrides any mismatch.



Fails closed

Three veto cycles without convergence → escalates to a human with the full audit trail. Never silent execution.



Foundry + Agent Framework

Both agents run live on Microsoft Foundry via the Microsoft Agent Framework. Offline reference brains prove the gate is model-agnostic.

The difference between an AI demo and a system you'd let run **unsupervised**.

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Propose → **Veto** → **Self-correct** → **Deliver**